



Ultra Compact Vision Sensor

Sensor head

IV-G500CA / IV-G500MA /
IV-G150MA /
IV-G300CA / IV-G600MA

Sensor amplifier

IV-G10 / IV-G15

Instruction Manual

Read this manual before using the product in order to achieve maximum performance.

Keep this manual in a safe place after reading it so that it can be used at any time.

For details of functions, refer to the IV/IV-G Series User's Manual (Monitor) or the IV/IV-G Series User's Manual (PC Software).

- The IV/IV-G Series User's Manual can be downloaded from the KEYENCE web site:
<http://www.keyence.com/>

Symbols

The following symbols alert you to important messages. Be sure to read these messages carefully.

	It indicates a hazardous situation which, if not avoided, will result in death or serious injury.
	It indicates a hazardous situation which, if not avoided, could result in death or serious injury.
	It indicates a hazardous situation which, if not avoided, could result in minor or moderate injury.
	It indicates a situation which, if not avoided, could result in product damage as well as property damage.

Important It indicates cautions and limitations that must be followed during operation.

Point It indicates additional information on proper operation.

Reference It indicates tips for better understanding or useful information.

Ethernet/IP is a trademark of ODVA, inc.

Cautions

- Unauthorized reproduction of this manual in whole or part is prohibited.
- The contents of this manual may be changed for improvements without prior notice.
- An utmost effort has been made to ensure the contents of this manual are as complete as possible. If there are any mistakes or questions, please contact a KEYENCE office listed in the back of the manual.
- Regardless of item (3), KEYENCE will not be liable for any effect resulting from the use of this unit.
- Any manuals with missing pages or other paging faults will be replaced.

The company names and product names used in this manual are registered trademarks or the trademarks of their respective companies.

Safety Information for IV-G Series

General Precautions

	- Do not use this product for the purpose to protect a human body or a part of human body. - This product is not intended for use as an explosion-proof product. Do not use this product in hazardous location and/or potentially explosive atmosphere.
	- You must verify that the IV-G Series are operating correctly in terms of functionality and performance before the start and the operation of the IV-G Series. - We recommend that you take substantial safety measures to avoid any damage in the event of a problem occurring.
	- KEYENCE never warrants the function or performance of the IV-G Series if it is used in manner that differs from the IV-G Series specifications contained in this instruction manual or if the IV-G Series are modified by yourself. - When the IV-G Series is used in combination with other instruments, functions and performance may be degraded, depending on operating conditions and the surrounding environment. - Do not place the instruments, including peripherals, under the rapid temperature change. It may cause condensation and may damage instruments or peripherals. - Remove the power cable of the sensor amplifier from the power supply if you do not use this product for a long time.

Important Instructions

Observe the following precautions to prevent malfunction of the IV-G Series and to ensure that it is used properly.

Precautions on use

	- The power of this product and instruments connected to this product must be turned off when the cable is to be installed or removed. Failure to do so may cause an electric shock or a product damage. - Use this product in the correct supply voltage. Failure to do so may cause a product damage.
	- For instructions - Do not turn OFF the power while setting the items or saving the settings. Otherwise, all or part of the setting data may be lost. - Do not let water, dust or oil stick to the camera/light of the sensor head. Failure to do so may cause a malfunction. - When this product becomes dirty, do not rub it with a wet cloth, benzene, thinner, or alcohol. Doing so may change the color or shape of the unit. - If the unit is heavily contaminated, disconnect all the cables including the power supply cable, wipe off the dirt with a cloth soaked with mild detergent, and then wipe with a soft dry cloth. - For external master image registration - If the external master image registration is to be performed frequently, set "Write ROM when using Ext. Master Save?" of the input option to "No" to protect the nonvolatile memory of the internal sensor. When the option is set to "Yes", the nonvolatile memory is guaranteed to write for 100,000 times. - For automatic focus function - Automatic focus function is used for adjusting the focusing position at the time of installation. Disabled during the operation. - Focusing position can be registered in each program. The program configurations are guaranteed to switch for 100,000 times. If the focusing position does not need to be changed in each program, set "Auto Focus Adjustment Position" to "Common" to extend its duration. - Do not apply shock or vibration during the focusing position adjustment. Failure to do so may cause a product damage.

Measures to be taken when an abnormality occurs

NOTICE	In the following cases, turn the power OFF immediately. Using the IV-G Series in an abnormal condition could cause fire, electric shock, or malfunction. Contact our office for repair. • If water or debris enters the IV-G Series. • If the IV-G Series is dropped or the case is damaged. • If abnormal smoke or odor emanates from the IV-G Series.

Precautions on installation

NOTICE	• To use this product correctly and safely, avoid installing it in the following locations. Failure to do so may cause fire, electric shock, or malfunction. • Outdoors • Altitude above 2000 m • Locations that are humid, dusty or poorly ventilated • Locations where the temperature is high such as those exposed to direct sunlight • Locations where there are flammable or corrosive gases • Locations where the unit may be directly subjected to vibration or impact • Locations where water, oil, or chemicals may splash onto the unit • To improve the anti-noise feature, install the unit following the precautions below. Otherwise, a malfunction may occur. • Ground the FG cable of the sensor amplifier. • Do not mount the unit in a cabinet where high-voltage equipment is already installed. • Mount the unit as far from power lines as possible. • Separate the unit as far as possible from the devices that emit strong electric or magnetic field (such as solenoid or chopper). • Separate the I/O signal line from the power line or high-voltage line. • For power supply • Noise superimposed on the power supply could cause malfunction. Use a stabilized DC power supply configured with an isolation transformer. • When using a commercially available switching regulator, be sure to ground the frame ground terminal. • Attach the sensor amplifier to the DIN rail then set it on the sheet metal. • Devices including this unit are precision components. Do not apply shock or vibration. • When connecting to a network, let engineers who are knowledgeable about networks handle it.

Safety Precautions on LED Product

The degree of risk of this product is shown below.

Model	Light source	Risk Group*
IV-G500CA	White LED	Risk Group 1
IV-G500MA		
IV-G150MA		
IV-G300CA	White LED	Exempt Group
IV-G600MA	Infrared LED	Exempt Group

* LED product is classified as shown below according to IEC 62471.

- Exempt Group Does not pose any photobiological hazard.
- Risk Group 1 (Low-Risk) Does not pose a hazard due to normal behavioral limitations on exposure.
- Risk Group 2 (Moderate-Risk) Does not pose a hazard due to the aversion response to very bright light sources or due to thermal discomfort.
- Risk Group 3 (High-Risk) May pose a hazard even for momentary or brief exposure.

Precautions on Regulations and Standards

UL Certification

This product is a UL/C-UL Listed product.

- UL File No. E301717
- Category NRKH2, NRKH8

Be sure to consider the following specifications when using this product as a UL/C-UL Listed Product.

- Use a power supply with Class 2 output defined in NFPA70 (NEC: National Electrical Code).
- Power supply/External input/Control output shall be connected to a single Class 2 source only.
- Use copper wire of AWG 16 to 18 (stranded wire) whose temperature rating is 60°C or higher for the power terminal block.
- Enclosure Type 1 (Based on UL50).
- Use under pollution level 2.

CE Marking

Keyence Corporation has confirmed that this product complies with the essential requirements of the applicable EU Directive, based on the following specifications. Be sure to consider the following specifications when using this product in the Member State of European Union.

● EMC Directive

- Applicable Standard EN60947-5-2, Class A
- The length of power I/O cable and Ethernet cable must be less than or equal to 30 m.

Remarks:

These specifications do not give any guarantee that the end-product with this product incorporated complies with the essential requirements of EMC Directive. The manufacturer of the end-product is solely responsible for the compliance on the end-product itself according to EMC Directive.

KC mark (Republic of Korea)

Class A device, Equipment name: Vision Sensor

This product is an industrial electromagnetic wave generating device (Class A) and is intended for use in all establishments other than domestic.

A 급 기기 (업무용 방송통신기자재), 기자재 명칭:

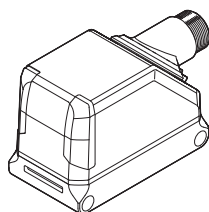
Vision Sensor

이 기기는 업무용 (A 급) 전자파적합기기로서 판매자또는 사용자는 이 점을 주의하시기 바라며, 가정외의 지역에서 사용하는 것을 목적으로 합니다.

Checking the Package Contents

Sensor head

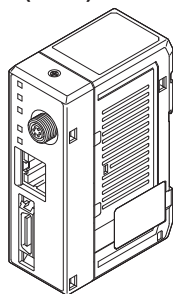
- IV-G500CA
- IV-G500MA
- IV-G150MA
- IV-G300CA
- IV-G600MA



Sensor head x 1

Sensor amplifier

- IV-G10 (Main)



Sensor amplifier (Main) x 1

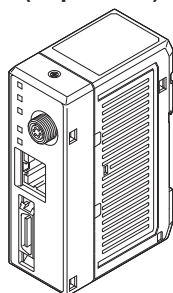
Instruction Manual x 1
(This manual)



LAN port cap (for Main)

Power terminal block
(mounted in the amplifier)

- IV-G15 (Expansion)



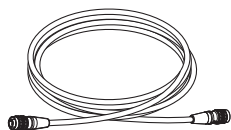
Sensor amplifier (Expansion) x 1

End unit x 2

Option

■ Sensor head/amplifier cable for IV-G series

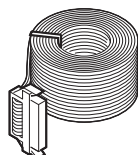
- OP-87903 (2 m)
- OP-87904 (5 m)
- OP-87905 (10 m)



Sensor head/amplifier cable for IV-G series x 1

■ I/O cable for IV-G series

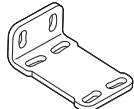
- OP-87906 (3 m)



I/O cable for IV-G series x 1

■ Vertical mounting bracket for IV-G series

- OP-87908



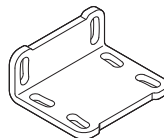
Mounting bracket x 1



Mounting screw (M4 x L8) x 2
* Double-washer sems screw

■ Transverse mounting bracket for IV-G series

- OP-87909



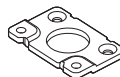
Mounting bracket x 1



Mounting screw (M4 x L8) x 2
* Double-washer sems screw

■ Adjustable bracket for IV-G series

- OP-87910



Mounting bracket x 1



Mounting screw (M4 x L8) x 2
* Double-washer sems screw



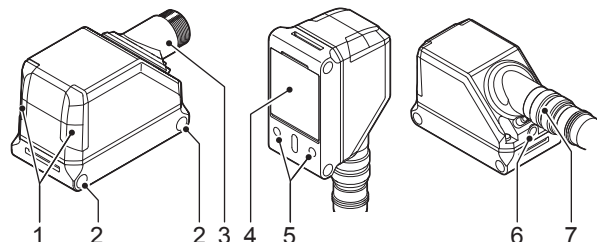
Bracket x 1

Reference For details on operations of the products below, refer to the instruction manual of each product.

- Dome attachment for IV-G series (Small) IV-GD05
- Dome attachment for IV-G series (Large) IV-GD10
- Polarizing filter attachment for IV-G500CA/500MA/150MA OP-87899
- Polarizing filter attachment for IV-G300CA OP-87900
- Polarizing filter attachment for IV-G600MA OP-87901
- Spread lens attachment for IV-G150MA OP-87902

Name and Function of Each Part

■ Name and function of each part of the sensor head



1 Indicator light

Indicates the operating status of the sensor head.

☐ "Operation of the sensor head indicator light" (Page 4)

2 Mounting part

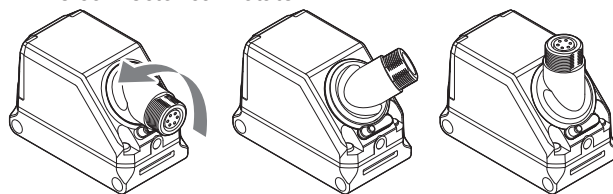
Used for mounting the sensor head.

Also used for mounting to the dome attachment for IV-G series.

☐ "Mounting the Sensor Head" (Page 5)

3 Connector for the sensor head/amplifier cable for IV-G series

The connector can rotate.



Example of rotating

4 Camera and built-in light

5 Mounting part for the spread lens attachment for IV-G150MA

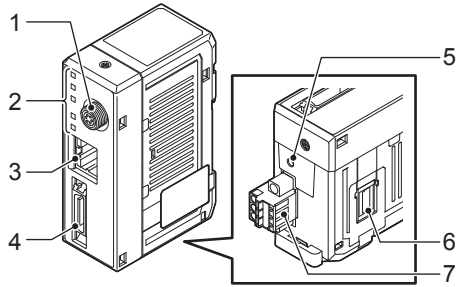
6 Mounting part for the polarizing filter attachment for IV-G500CA/500MA/150MA, polarizing filter attachment for IV-G300CA, polarizing filter attachment for IV-G600MA

7 Connector for the sensor head/amplifier cable for IV-G series

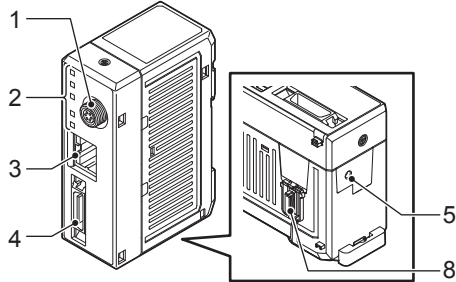
☐ "Cables" (Page 7)

■ Name and function of each part of the sensor amplifier

IV-G10



IV-G15



1 Connector for the sensor head/amplifier cable for IV-G series

☐ "Cables" (Page 7)

2 Indicator light

Indicates the operating status of the sensor amplifier.

☐ "Operation of the sensor amplifier indicator light" (Page 4)

3 Ethernet connector

Use for connecting with the intelligent monitor (IV-M30) or PC. For details of connections, refer to the IV/IV-G Series User's Manual (Monitor) or the IV/IV-G Series User's Manual (PC Software).

When the connector is not used, attach the included LAN port cap.

4 I/O connector

Used for connecting the I/O cable for IV-G series (OP-87906).

5 IP reset switch

Used for resetting the IP address assigned to this sensor amplifier.

For details, refer to the IV/IV-G Series User's Manual (Monitor) or the IV/IV-G Series User's Manual (PC Software).

6 Add-on connector

Used for adding the IV-G15 (Expansion).

☐ "Mounting the Sensor Amplifier" (Page 6)
Protection sheet is attached for shipping.

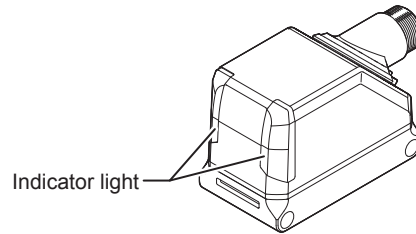
7 Power terminal block

Supplies power to the IV-G10. This is not equipped with the IV-G15.

8 Add-on connector

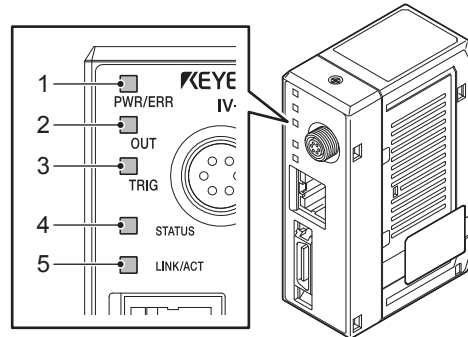
Used for connecting the IV-G10 (Main).

■ Operation of the sensor head indicator light



- Green (ON) Comprehensive result is "OK".
- Green (Blink) Under startup or setting in progress. Operation is stopped. Blinks once a second.
- Red (ON) Comprehensive result is "NG".
- Red (Blink) An error has occurred.
- (OFF) Standby status until the first judgment finishes after starting the operation or after switching the program number. Version of the sensor head and sensor amplifier does not match.
- Green and red ... "LED Blinking" is requested. are blinking alternately.

■ Operation of the sensor amplifier indicator light



1 PWR/ERR

- Green (ON) Operating.
- Green (Blink) Setting processing. Operation is stopped. Blinks once a second.
- Red (ON) Unrecoverable error has occurred.
- Red (Blink) Recoverable error has occurred.
- (OFF) Power is not supplied.

2 OUT

Indicates the comprehensive result.

- Green Comprehensive result is "OK".
- Red Comprehensive result is "NG".
- (OFF) Setting processing. Standby status until the first judge finishes after starting the operation or after switching the program number.

3 TRIG

Green light lights up (one-shot) according to input of the internal or external trigger.

4 STATUS

Indicates the connection status within the monitor.

- Green (ON) Normally connected with monitor or PC.
- Green (Blink) IP address has been retrieved but the sensor amplifier is not correctly connected with monitor or PC.
- Red (ON) The IP address coincides with other device or a network loop is detected.
- (OFF) IP address is not assigned. Sensor amplifier is not correctly connected with monitor or PC.

5 LINK/ACT

Indicates the linking status within monitor or Ethernet switch.

- Green (ON) Normally linked.
- Green (Blink) Normally linked, and the data is sending/receiving.
- (OFF) Sensor amplifier is not normally linked.

Mounting the Sensor Head

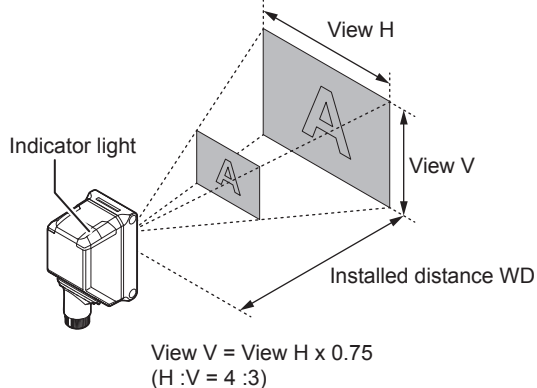
NOTICE	Do not place the sensor in the environment that exceeds the limit of sensor's resistance to the environment, or environment that propagates the vibration directly to the sensor. Those may cause a damage or malfunction.
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Point

- View and optical axis have individual differences. Adjust the position by checking the actual image at the time of installation.
- Place the sensor where no ambient light has effect. Ambient light includes solar light, lights of other devices, and photoelectric sensors. Also, be careful when the light intensity of the ambient light changes. Use the shield to protect when the location cannot be changed.
- Place the sensor where no object can block out the internal light or the view of the sensor.
- Detection may become unstable due to the lights if multiple sensors are placed nearby each other. Take measures as described below.
 - Use the mutual interference prevention function.
 - Delay the timing of external trigger inputs.
 - Use the shield to avoid interference.

Checking the view and installed distance

For IV-G Series, installed distance between the view and target is different depending on the type of the sensor head. Check the type of the sensor head to be used and its view, and place it in the proper distance.



● IV-G150MA

Installed distance WD (mm)	View H (mm)	View V (mm)
40	8	6
50	11	8
100	22	16
150	34	24

● IV-G500CA / IV-G500MA

Installed distance WD (mm)	View H (mm)	View V (mm)
20	10	7.5
50	20	15
100	40	30
200	80	60
300	120	90
400	160	120
500	200	150

● IV-G300CA

Installed distance WD (mm)	View H (mm)	View V (mm)
40	42	31
100	95	71
150	140	105
200	185	139
250	230	172
300	275	206

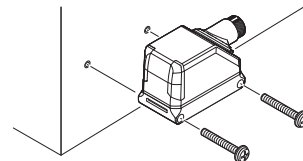
● IV-G600MA

Installed distance WD (mm)	View H (mm)	View V (mm)
40	42	31
100	95	72
200	185	140
300	275	208
400	365	276
500	455	344
600	550	412

Mounting the Sensor Head

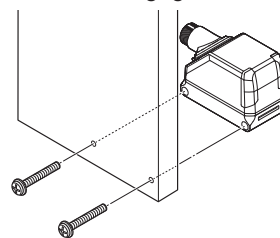
■ When screws are secured from the sensor head

- Screw : M3 x 2
- Tightening torque : 0.3 to 0.6N·m



■ When screws are secured from the wall

- Screw : M4 x 2
- Tightening torque : 0.7 to 1.5N·m
- Screw hole on the sensor head: M4 female (whose length of screw engagement should be 2.5 to 3.5 mm)



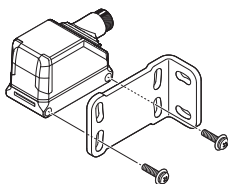
Attaching the optional mounting bracket

■ When using the vertical mounting bracket for IV-G series (OP-87908)

Attach the bracket using the screws attached to OP-87908.

- Tightening torque : 0.7 to 1.5N·m

Mounting examples

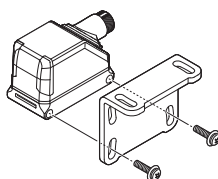


■ When using the transverse mounting bracket for IV-G series (OP-87909)

Attach the bracket using the screws attached to OP-87909.

- Tightening torque : 0.7 to 1.5N·m

Mounting examples

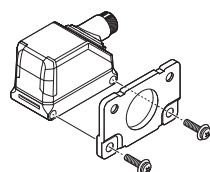


■ When using the adjustable bracket for IV-G series (OP-87910)

Attach the bracket using the screws attached to OP-87910.

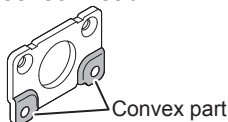
- Tightening torque : 0.7 to 1.5N·m

Mounting examples



Point

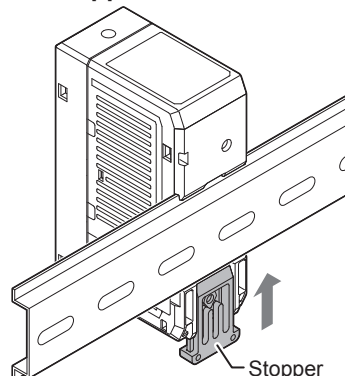
When using the adjustable bracket for IV-G series (OP-87910), abut the convex part to the sensor head.



Mounting the Sensor Amplifier

Mounting the IV-G10 (Main)

- 1 Adjust the position of the DIN mounting adapter and the DIN rail by pulling down the stopper of the sensor amplifier IV-G10 (Main), and then push up the stopper to lock it.



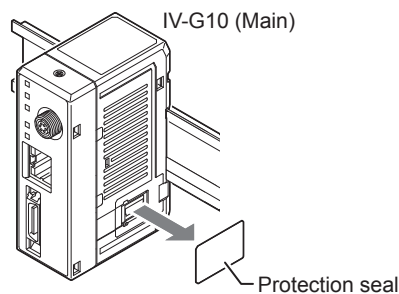
Mounting the IV-G15 (Expansion)

An expansion unit is used for attaching to the main unit. One expansion unit can be added to one main unit.

NOTICE

- For adding an expansion unit, be sure to mount it to the DIN rail.
- When a sensor amplifier (Expansion) is to be added, confirm that the power is turned OFF. If the sensor amplifier is added with the power ON, the unit may be damaged.
- Be sure to push the sensor amplifier (Expansion) into the main unit all the way. If the sensor amplifier is connected at an angle or is not pushed into the main unit firmly, the unit may be damaged.

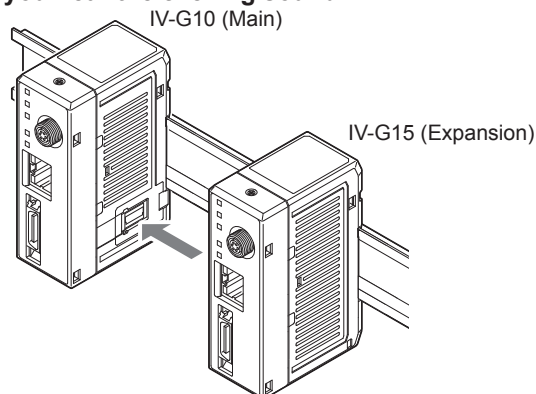
- 1 Remove the protection seal from the sensor amplifier IV-G10 (Main).



- 2 Mount the sensor amplifier IV-G15 (Expansion added) to the DIN rail.

Attach it in the same way as "Mounting the IV-G10 (Main)" (Page 6).

- 3 Press the expansion unit to the main unit until you hear the clicking sound.



- 4 Attach the end units (OP-26751: 2 pack) on both sides of the main and expansion units mounted, and secure them with two screws on each side.

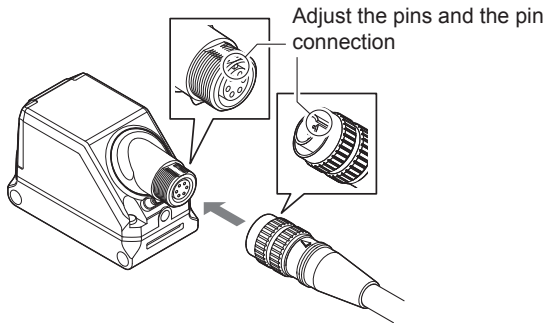
NOTICE

Attach the sensor amplifier to the DIN rail then set it on the sheet metal.

Cables

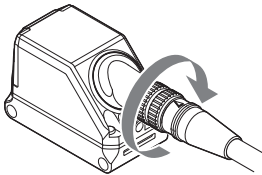
Connecting the sensor head/amplifier cable for IV-G series and sensor head

- 1 Insert the cable into the sensor as shown below.



- 2 Turn the screw-on connector in the clockwise direction to tighten it.

• Tightening torque : 0.8N·m or less

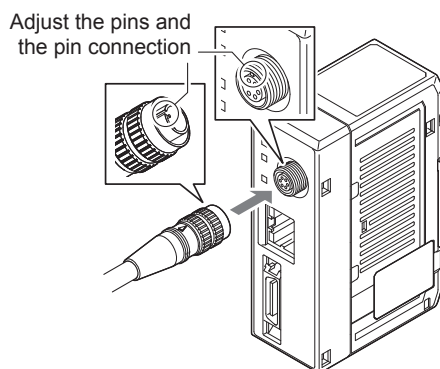


NOTICE

To tighten the screw by hand, turn it until the connector of the cable touches the sensor head firmly. Then turn it more about by 5° using a tool such as pliers.

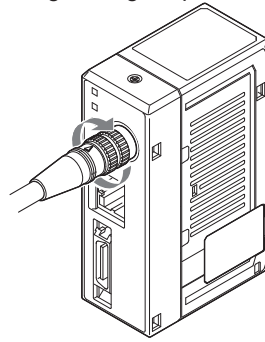
Connecting the sensor head/amplifier cable for IV-G series and sensor amplifier

- 1 Insert the cable into the sensor as shown below.

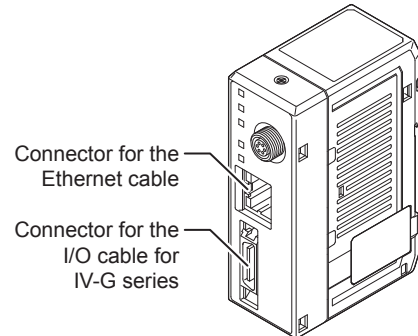


- 2 Turn the screw-on connector in the clockwise direction to tighten it.

• Tightening torque : 0.4 to 0.8N·m

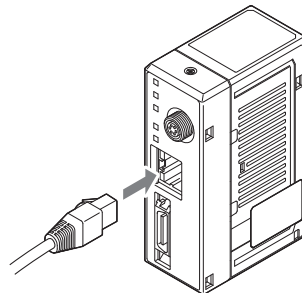


Connecting the cable to the sensor amplifier

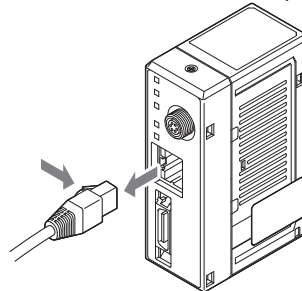


■ Connecting the Ethernet cable or LAN cable

Connect the cable to the sensor amplifier as shown below.

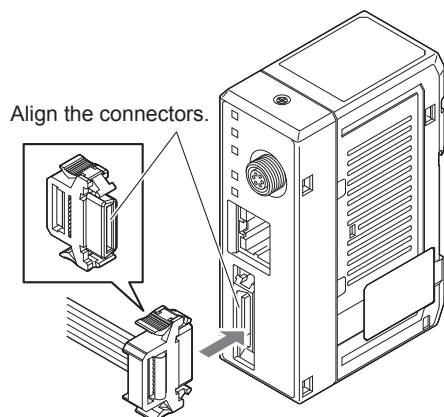


To disconnect the cable, pull the cable while holding the tab on the Ethernet cable pushed in.

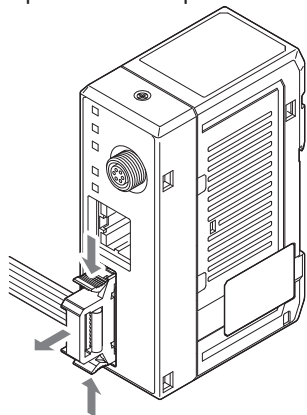


■ Connecting the I/O cable for IV-G series (OP-87906)

Connect the cable to the sensor amplifier as shown below.



To disconnect the cable, hold the tabs on both sides of the connector pushed in and pull it out.



● Terminal number and wiring color of the I/O cable for IV-G series (OP-87906)

Terminal No.	Wiring color	Name	Assigning default value	Description
A1	Brown	IN1	Ext. Trigger ↑	Set external trigger. Rising timing (↑) or falling timing (↓) can be set.
A2	Red	IN2	OFF	Input assignable function • Program bit0 to bit4 • Clear Error • Ext. Master Save • OFF (not used)
A3	Orange	IN3	OFF	
A4	Yellow	IN4	OFF	
A5	Green	IN5	OFF	
A6	Blue	IN6	OFF	
A7	Purple	Unused	Unused	Unused
A8	Gray	Unused	Unused	
A9	White	Unused	Unused	
A10	Black	Unused	Unused	

Terminal No.	Wiring color	Name	Assigning default value	Description
B1	Brown	OUT1	Total Status (N.O.)	Output assignable function • Total Status • Total Status NG • RUN • BUSY • Error • Position Adjustment • Status result of each tool (Too 01 to 16) • Logical operation result of each tool (Logic 1 to 4) • OFF (not used)
B2	Red	OUT2	BUSY (N.O.)	
B3	Orange	OUT3	Error (N.C.)	
B4	Yellow	OUT4	OFF	
B5	Green	OUT5	OFF	
B6	Blue	OUT6	OFF	
B7	Purple	OUT7	OFF	
B8	Gray	OUT8	OFF	
B9	White	Unused	Unused	Unused
B10	Black	Unused	Unused	

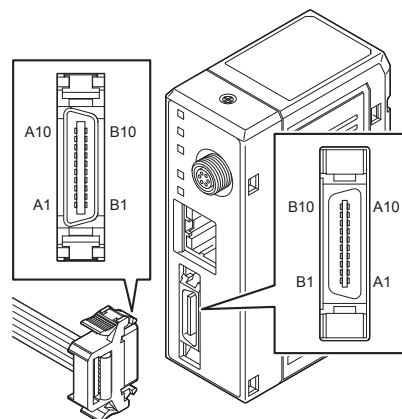
Cable specification : AWG28

Reference The output assignment, N.O./N.C., and input line assignment can be changed. For details, refer to the IV/IV-G Series User's Manual (Monitor) or the IV/IV-G Series User's Manual (PC Software).

NOTICE

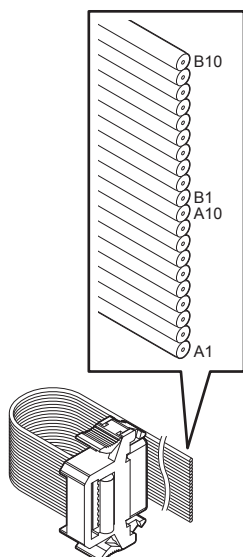
- Individually insulate the non-used input-output cables.
- For input cables of this sensor, connect with non-contact output (transistor output/SSR output). For contact output (relay output), incorrect input may be operated due to the contact bouncing.

● Pin assignment (I/O cable for IV-G series: connector side)



Pin No.	Color	Pin No.	Color
B10	Black	A10	Black
B9	White	A9	White
B8	Gray	A8	Gray
B7	Purple	A7	Purple
B6	Blue	A6	Blue
B5	Green	A5	Green
B4	Yellow	A4	Yellow
B3	Orange	A3	Orange
B2	Red	A2	Red
B1	Brown	A1	Brown

● **Pin assignment**
(I/O cable for IV-G series: connector of the external device)



Pin No.	Color
B10	Black
B9	White
B8	Gray
B7	Purple
B6	Blue
B5	Green
B4	Yellow
B3	Orange
B2	Red
B1	Brown
A10	Black
A9	White
A8	Gray
A7	Purple
A6	Blue
A5	Green
A4	Yellow
A3	Orange
A2	Red
A1	Brown

■ **Supplying power to the sensor amplifier**

The power terminal block is attached on the bottom of the sensor amplifier IV-G10 (Main).

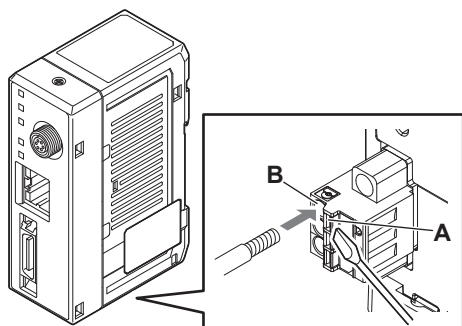
- 24 V : Supplies 24 V DC
- 0 V : Wire to 0 V of the power supply.
- FG : Ground

NOTICE

Follow the instructions given below to avoid damage caused by bad connections.

- The nominal cross-sectional area of the wire for connecting the power should be 0.8mm^2 to 1.3mm^2 (AWG 16 to 18).
- Length of removing insulation from the wire should be about 9 mm.
- The stripped part of the wire should not be soldered (pre-soldered).
- Insert the wires into the power terminal block all the way.
- Connect the wires directly. Do not use a crimp-type terminal.

1 Insert the wire to B by pushing A in with a screwdriver.



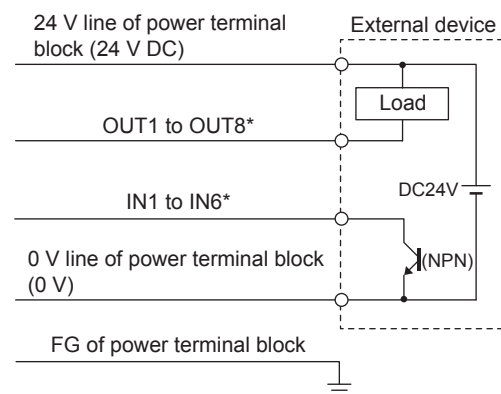
2 When the wire is inserted into the end, remove the screwdriver.

Pull the wire out slightly to confirm that it is secured.

■ **Connection diagram**

● **Selecting NPN output**

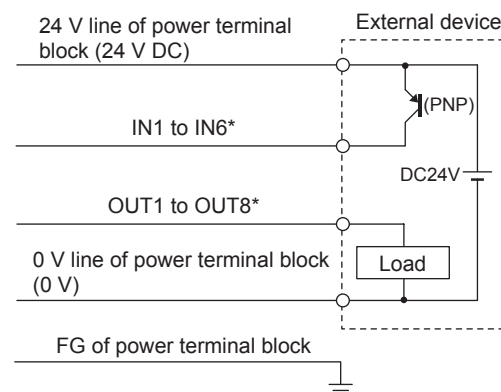
When NPN is selected in I/O format



- * For the terminal number and wiring color, refer to "●Terminal number and wiring color of the I/O cable for IV-G series (OP-87906)" (Page 8).

● **Selecting PNP output**

When PNP is selected in I/O format



- * For the terminal number and wiring color, refer to "●Terminal number and wiring color of the I/O cable for IV-G series (OP-87906)" (Page 8).

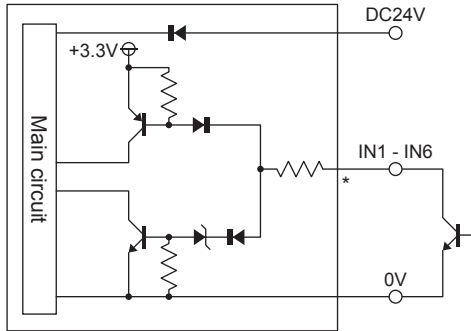
Input circuit

■ No-voltage input

(When the NPN output is selected)

When NPN is selected in I/O format, the circuit becomes no-voltage input circuit. External power supply is not necessary.

- ON voltage : 2 V or lower
- OFF current : 0.1 mA or lower
- ON current : 2 mA (short circuit)



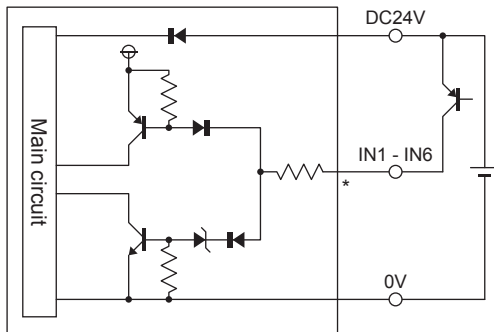
* Refer to "•Terminal number and wiring color of the I/O cable for IV-G series (OP-87906)" (Page 8).
Use by assigning the optional functions to IN2 to 6

■ Voltage input

(When the PNP output is selected)

When PNP is selected in I/O format, the circuit becomes voltage input circuit.

- Input maximum rating : 26.4 V
- ON voltage : 18 V or higher
- ON current : 2 mA (for 24 V)
- OFF current : 0.2 mA or lower



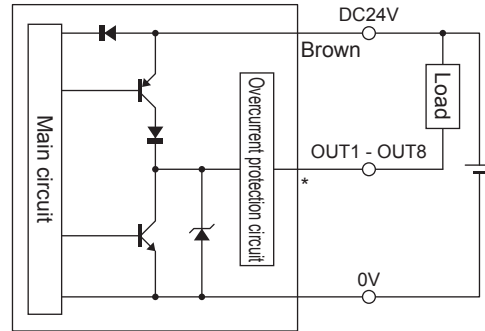
* Refer to "•Terminal number and wiring color of the I/O cable for IV-G series (OP-87906)" (Page 8).
Use by assigning the optional functions to IN2 to 6

Output circuit

■ When the NPN output is selected

When NPN is selected in I/O format, the circuit becomes open collector NPN output circuit.

- Maximum rating : 26.4 V, 50 mA
(20 mA when the IV-G15 (Expansion) is connected)
- Remaining voltage : 1.5 V or lower

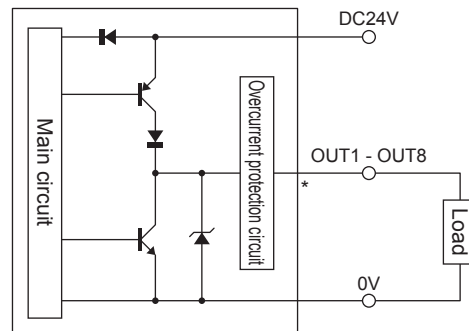


* Refer to "•Terminal number and wiring color of the I/O cable for IV-G series (OP-87906)" (Page 8).
Use by assigning the optional functions to OUT1 to OUT8

■ When the PNP output is selected

When PNP is selected in I/O format, the circuit becomes open collector PNP output circuit.

- Remaining voltage : 1.5 V or lower
- Maximum rating : 26.4 V, 50 mA
(20 mA when the IV-G15 (Expansion) is connected)
- Remaining voltage : 2 V or lower



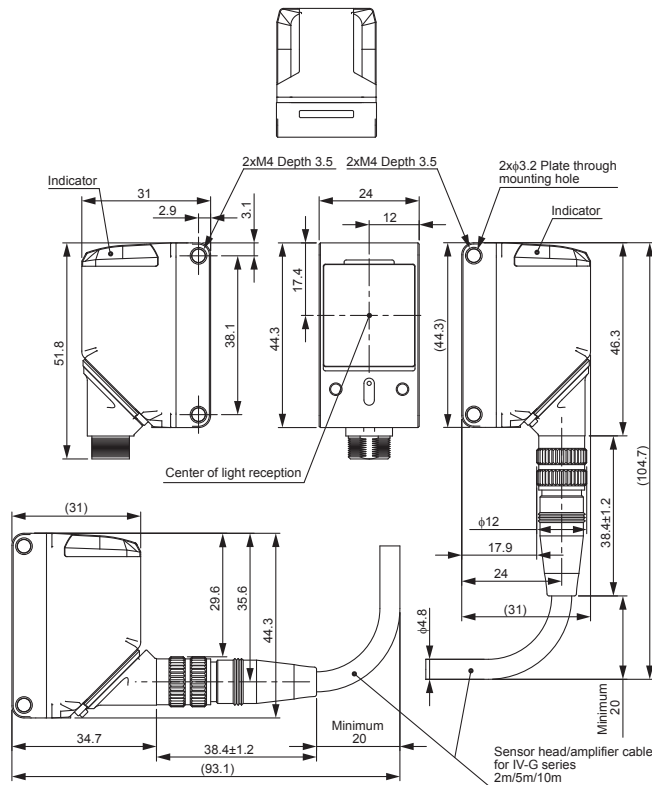
* Refer to "•Terminal number and wiring color of the I/O cable for IV-G series (OP-87906)" (Page 8).
Use by assigning the optional functions to OUT1 to OUT8

Dimensions

For detail of the dimensions, refer to "IV/IV-G Series User's Manual". The IV/IV-G Series User's Manual can be downloaded from the KEYENCE web site: <http://www.keyence.com/>

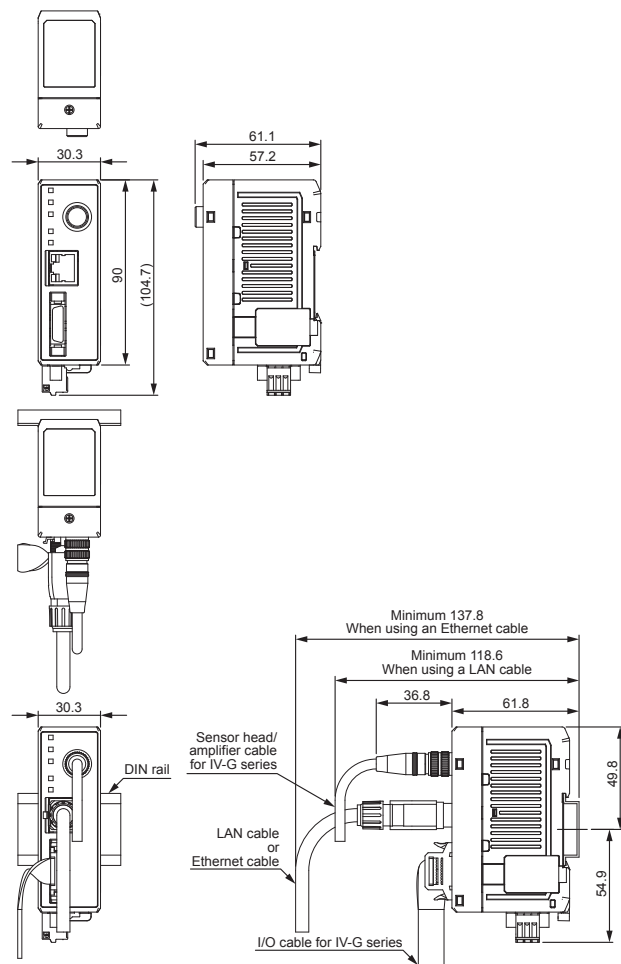
Sensor Head

IV-G500CA / IV-G500MA / IV-G150MA / IV-G300CA / IV-G600MA

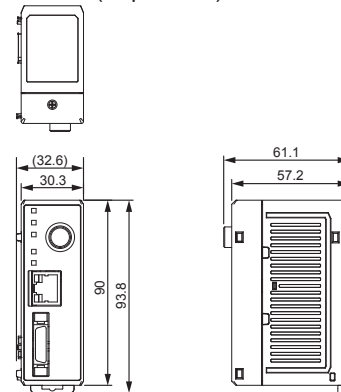


Sensor Amplifier

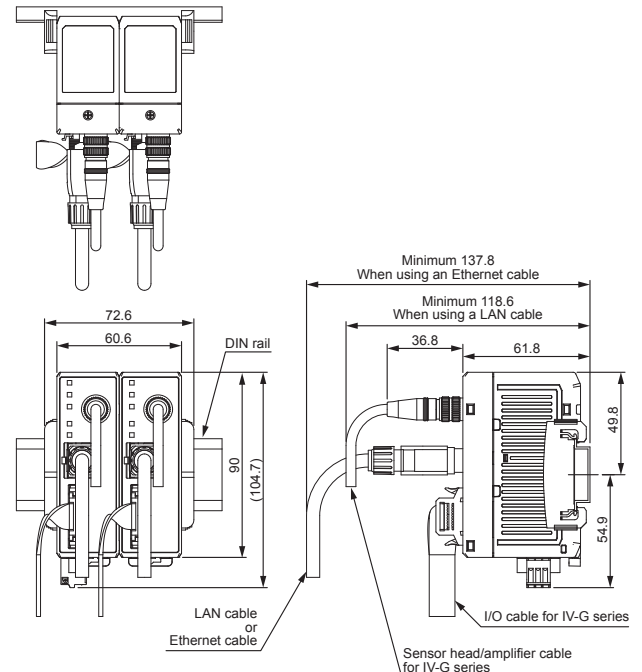
IV-G10 (Main)



IV-G15 (Expansion)

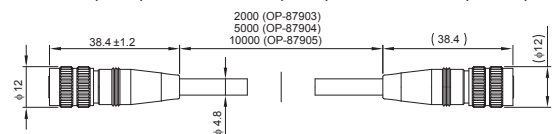


IV-G10 + IV-G15 + End Unit



Sensor head/amplifier cable for IV-G series

OP-87903 (2 m)/ OP-87904 (5 m)/ OP-87905 (10 m)



Specifications

Sensor Head

Model		IV-G500CA	IV-G500MA	IV-G150MA	IV-G300CA	IV-G600MA
Type		Standard type		Narrow view type	Wide view type	
Installed distance		20 to 500 mm		40 to 150 mm	40 to 300 mm	40 to 600 mm
View		Installed distance 20 mm : 10 (H) x 7.5 (V) mm to Installed distance 500 mm : 200 (H) x 150 (V) mm		Installed distance 40 mm : 8 (H) x 6 (V) mm to Installed distance 150 mm : 32 (H) x 24 (V) mm	Installed distance 40 mm : 42 (H) x 31 (V) mm to Installed distance 300 mm : 275 (H) x 206 (V) mm	Installed distance 40 mm : 42 (H) x 31 (V) mm to Installed distance 600 mm : 550 (H) x 412 (V) mm
Image sensor		1/3 inch color CMOS	1/3 inch monochrome CMOS	1/3 inch monochrome CMOS	1/3 inch color CMOS	1/3 inch monochrome CMOS
	Pixel	752 (H) x 480 (V)				
Focus Adjustment		Auto* ¹				
Exposure Time		1/10 to 1/50,000		1/20 to 1/50,000	1/25 to 1/50,000	1/50 to 1/50,000
Lighting	Amplifier type	White LED				Infrared LED
	Lighting method	Pulse lighting/DC lighting is switchable			Pulse lighting	
Indicators		2 (indicate the same status)				
Environmental resistance	Ambient temperature	0 to +50°C (No freezing)				
	Relative humidity	35 to 85 % RH (No condensation)				
	Vibration* ²	10 to 55 Hz, 1.5 mm double amplitude, 2 hours each for X, Y, and Z axes				
	Shock resistance* ²	500 m/s ² 6 different directions in 3 times				
	Enclosure rating* ³	IP67				
Material		Body: zinc diecast, front cover: acrylic (hard coating), indicator cover: TPU				
Weight		Approx. 75g				

Sensor Amplifier

Model		IV-G10 (Main)	IV-G15 (Expansion)
Tool	Type	Outline, Color area ^{*4} , Area ^{*5} , Position adjustment	
	Number ^{*6}	Detection tools : 16 tools, Position adjustment tool : 1 tool	
Switch settings (programs)		32 programs	
Image history ^{*7}	Number of cases	Color-type head: 100 images ^{*8} , monochrome-type head: 300 images ^{*9}	
	Condition	NG only/All is selectable	
Analysis information ^{*10}		OFF/Statistics/Histogram is switchable Statistics : Processing time (latest value, MAX, MIN, AVE), numbers of OKs, numbers of NGs, trigger numbers, trigger errors, judge results list by tools Histogram : Histogram, matching degree (latest value, MAX, MIN, AVE), numbers of OKs, numbers of NGs	
Other functions		HDR, High Gain, Color Filters ^{*4} , Digital Zoom (x2, x4) ^{*11} , Brightness Correction, Tilt correction, White balance ^{*4} , Mask outline, Mask area, Test run, ToolAutoTune, Input monitor, Output test, Security settings, Simulator, Prevention function for mutual interference among heads, Total status output, Direct connection (2 units or more), List of NG Sensor Occurrences, NG Hold Function	
Indicators		PWR/ERR, OUT, TRIG, STATUS, LINK/ACT	
Input		No-voltage input/voltage input is switchable For no-voltage input : ON voltage 2 V or lower, OFF current 0.1 mA or lower, ON current 2 mA (short circuit) For voltage input : Maximum input rating 26.4 V, ON voltage 18 V or higher, OFF current 0.2 mA or lower, ON current 2 mA (for 24 V)	
	Inputs	6 inputs (IN1 to IN6)	
	Function	IN1 : External trigger, IN2 to IN6 : Enable by assigning the optional functions Assignable functions : Program switching, Clear error, External master image registration, Main/Expansion simultaneous input	

Model		IV-G10 (Main)	IV-G15 (Expansion)
Output		Open collector output NPN/PNP is switchable, N.O./N.C. is switchable For open collector NPN output : Maximum rating 26.4 V 50 mA (20 mA when an Expansion unit (IV-G15) is connected), remaining voltage 1.5 V or lower For open collector PNP output : Maximum rating 26.4 V 50 mA (20 mA when an Expansion unit (IV-G15) is connected), remaining voltage 2 V or lower	
	Outputs	8 outputs (OUT1 to OUT8)	
	Function	Enable by assigning the optional functions Assignable functions : Total judge result, RUN, BUSY, Error, Position adjustment result, Judge result of each tool, Logical operation result of each tool, Main/Expansion logic output	
Ethernet ^{*12}	Standard	100BASE-TX/10BASE-T	
	Connector	RJ-45 8pin connector	
Network function		FTP client, EtherNet/IP, PROFINET	
Rating	Power voltage	DC 24 V $\pm$ 10% (including ripple)	Supplied from the Main unit
	Consumption current	0.8 A or less, 1.5 A or less when an Expansion unit (IV-G15) is connected (excluding output load)	
Environmental resistance	Ambient temperature	0 to +50°C (No freezing) ^{*13}	
	Relative humidity	35 to 85 % RH (No condensation)	
Material		Body: Polycarbonate	
Weight		Approx. 150 g	

- *1 The focusing position can be automatically adjusted at the time of installation. Disabled during the operation. The focusing position can be registered by programs.
- *2 Except when the dome attachment (IV-GD05/IV-GD10) is mounted.
- *3 Except when the polarizing filter attachment (OP-87899/OP-87900/OP-87901/OP-87902) is mounted.
- *4 Color type only.
- *5 Monochrome type only.
- *6 Tools can be installed by programs.
- *7 Saves to the memory in the sensor. The images saved in the sensor can be backed up to the USB memory installed to the intelligent monitor (IV-M30) or to the PC by the PC Software for IV/IV-G series (IV-H1).
- *8 70 images when using the FTP client function.
- *9 210 images when using the FTP client function.
- *10 This can be displayed on the intelligent monitor (IV-M30) or by the PC Software for IV/IV-G series (IV-H1).
- *11 Color type and monochrome type.
- *12 This is for connection with the intelligent monitor (IV-M30) or the PC Software for IV/IV-G Series(IV-H1).
- *13 Mount to the DIN rail, then set them on the sheet metal.

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KEYENCE CORPORATION

1-3-14, Higashi-Nakajima, Higashi-Yodogawa-ku,
Osaka, 533-8555, Japan
PHONE: +81-6-6379-2211

www.keyence.com

AUSTRIA Ph: +43 2236 378266 0	HONG KONG Ph: +852-3104-1010	NETHERLANDS Ph: +31 40 20 66 100	TAIWAN Ph: +886-2-2721-8080
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A6WW1-MAN-1109

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13286GB 1109-3 [96M13286] Printed in Japan

